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Inventor(s)

Takemoto; Yoshiyuki et al.

### LURE

#### Abstract

A lure **1** includes a body **3**, a support **5**, and a weight **7**. The body **3** includes a head **13a** in its front portion and a tail **13b** in its rear portion. The support **5** extends in the front-rear direction inside the body **3**. The weight **7** is supported by the support **5** so as to be movable in the front-rear direction. At least part of the weight **7** is movable in a transverse direction that intersects with the front-rear direction.

**Inventors:** Takemoto; Yoshiyuki (Sakai City, Osaka, JP), Nakamura; Kazuto (Sakai City, Osaka, JP)

**Applicant:** Shimano Inc. (Sakai City, JP)

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#### Background/Summary

## CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Japanese Patent Application No. 2024-35194, filed Mar. 7, 2024, and Japanese Patent Application No. 2024-35195 filed Mar. 7, 2024. The contents of those applications are incorporated by reference herein in their entirety.

## TECHNICAL FIELD

[0002] The claimed invention relates to lures.

## BACKGROUND ART

[0003] The lure described in Japanese Patent Publication No. 2021-193921 includes a body, a shaft, a weighted element, and an elastic member. The shaft is comprised of straight wire material. The shaft extends linearly in the front-rear direction inside the body. The weighted element is supported by a guide member inside the body so as to be movable in the front-rear direction. The elastic member urges the weighted element forward.

[0004] In this type of lure, when the lure is cast, the weighted element moves linearly toward the tail of the body along the shaft by inertia while compressing the elastic member. When the lure hits the water, the weighted element is pushed linearly back toward the head of the body along the shaft by the urging force of the elastic member. The movement of the weighted element suitably maintains the posture of the lure when the lure is cast and the posture when it hits the water. While the lure is moving through the water, the weighted element moves linearly in the front-rear direction along the shaft in response to an external force applied to the lure.

## BRIEF SUMMARY

[0005] In the lure described in Japanese Patent Publication No. 2021-193921, the shaft is linear in shape. In this case, the movement path of the weighted element is restricted to a linear trajectory. Thus, depending on the shape and internal structure of the lure, it might be difficult to place the weighted element in an optimal position. In other words, the degree of freedom of the weighted element's movement path is limited to a straight line, thereby making it difficult to place the center of gravity of the lure at a position the designer desires.

[0006] The lure described in Japanese Patent No. 5530677 includes a body, a guide wire, a first weighted element, a second weighted element heavier than the first weighted element, and a step portion. The guide wire extends linearly in the front-rear direction inside the body. The first weighted element is positioned behind the second weighted element inside the body. The second weighted element is supported by the guide wire inside the body so as to be movable in the front-rear direction.

[0007] In this type of lure, when the lure is cast, the first and second weighted elements move linearly along the guide wire toward the tail of the body due to inertial force. When the lure hits the water, the first and second weighted elements move linearly along the guide wire toward the head of the body. While the lure is moving through the water, the first weighted element is positioned between the second weighted element and the step portion, thereby restricting the second weighted element from moving backward during movement through the water.

[0008] In the conventional lure, the first weighted element is positioned between the second weighted element and the step portion, so that the center of gravity of the lure is adjusted toward the head while the lure is moving through the water. In this case, the first weighted element and the second weighted element move independently. As a result, there is a risk that, when the second weighted element moves toward the head, the first weighted element may not follow it and may not be positioned between the second weighted element and the step portion. In addition, to prohibit the first weighted element from exceeding the step portion while the lure is moving through the water, the design is required to adjust the weight and diameter of the first weighted element through trial and error.

[0009] The claimed invention aims to provide a lure whose center of gravity can be adjusted appropriately. The claimed invention also aims to provide a lure whose center of gravity can be

stabilized while moving through the water.

[0010] In a first aspect of the claimed invention, the lure includes a body, a support, and a weight. The body has a head in its front portion and a tail in its rear portion. The support extends in the front-rear direction inside the body. The weight is supported by the support so as to be movable in the front-rear direction. At least part of the weight is movable in a transverse direction that intersects the front-rear direction.

[0011] In this lure, because the weight is movable in the transverse direction, the degree of freedom in the weight's movement path can be enhanced compared to the prior art. Consequently, the center of gravity of the lure can be appropriately adjusted. Furthermore, the center of gravity while the lure is moving through the water can be stabilized with a simple configuration.

[0012] In a second aspect of the claimed invention, the lure according to the first aspect may be configured as follows: The support is supported by the body so as to be movable in the transverse direction.

[0013] In this lure, the support is movable in the transverse direction, and the weight is supported by the support so as to be movable in the front-rear direction. As a result, when the support moves in the transverse direction, the weight also moves in the transverse direction together with the support. That is, the center of gravity of the lure can be appropriately adjusted. Furthermore, the weight moves both in the front-rear direction and the transverse direction in response to the action (e.g., winding operation) performed by the angler, and thereby allowing the lure to behave irregularly.

[0014] In a third aspect of the claimed invention, the lure according to the second aspect may be configured as follows: The support is an axial member with a shaft-like shape. The axial member is able to oscillate around the tail side. With this configuration, oscillation of the weight supported by the support enhances the degree of freedom of the movement path of the weight.

[0015] In a fourth aspect of the claimed invention, the lure according to the third aspect may be configured as follows: The axial member includes a front-end portion disposed on the head side and a rear-end portion disposed on the tail side and supported by the body to be able to oscillate. The body has a groove for guiding the front-end portion in the transverse direction. With this configuration, the front-end portion of the axial member is guided in the transverse direction by the groove of the body, so that the weight supported by the support can move stably in the transverse direction.

[0016] In a fifth aspect of the claimed invention, the lure according to the second aspect may further include an urging member. The urging member urges the weight at least forward. This configuration facilitates returning of the weight toward the head.

[0017] In a sixth aspect of the claimed invention, the lure according to the first aspect may further include a restricting portion. The restricting portion is configured to limit the weight from moving rearward. The restricting portion is positioned on an inner surface of the body in the front portion of the body. The weight is restricted by the restricting portion when at least part of the weight moves in the transverse direction.

[0018] In this lure, the weight is restricted by the restricting portion as it moves at least partially in the transverse direction. Consequently, in this lure, the restricting portion in the front portion of the body easily restricts the weight from moving rearward. In other words, the center of gravity of the lure can be stabilized with a simple structure while the lure is moving through the water.

[0019] In a seventh aspect of the claimed invention, the lure according to the sixth aspect may be configured as follows: The support is supported by the body so as to be movable in the transverse direction. When the support moves in the transverse direction, the weight is restricted by the restricting portion. This configuration allows the weight to engage with the restricting portion, which reliably restricts the weight from moving rearward.

[0020] In an eighth aspect of the claimed invention, the lure according to the seventh aspect may be configured as follows: The restricting portion is a recess formed on the inner surface of the body.

With this configuration, the engagement of the weight with the recess ensures the restriction of rearward movement of the weight.

[0021] In a ninth aspect of the claimed invention, the lure according to the eighth aspect may further include a magnet. The magnet is positioned on the body so as to face the recess with the support in between. The weight is at least partially comprised of magnetic material. With this configuration, the engagement between the weight and the recess can be easily released by the magnetic force of the magnet. Additionally, the disengagement state between the weight and the recess can be suitably maintained.

[0022] In a tenth aspect of the claimed invention, the lure according to the ninth aspect may be configured as follows: The support is an axial member with a shaft-like shape. The axial member is able to oscillate around the tail side. This configuration facilitates the movement of the weight toward the magnet.

[0023] In an eleventh aspect of the claimed invention, the lure according to the seventh or eighth aspect may further include an urging member. The urging member urges the weight at least forward. This configuration facilitates returning of the weight toward the head.

[0024] In a twelfth aspect of the claimed invention, the lure according to the sixth aspect may be configured as follows: The weight includes a weight body and a sliding portion. The restricting portion includes a slope that inclines toward an outer surface of the body. When at least part of the weight moves in the transverse direction, the sliding portion comes into contact with the slope, and the slope restricts the weight from moving rearward through sliding resistance against the sliding portion. With this configuration, the contact between the sliding portion of the weight and the slope of the restricting portion easily restricts the weight from moving rearward.

[0025] In a thirteenth aspect of the claimed invention, the lure according to the twelfth aspect may be configured as follows: The support is an axial member with a shaft-like shape. The sliding portion includes a main body and a hole in the main body through which the axial member is inserted. The hole has an inner diameter greater than the outer diameter of the axial member. With this configuration, the greater inner diameter of the hole relative to the outer diameter of the axial member facilitates the movement of the sliding portion in the transverse direction.

[0026] In a fourteenth aspect of the claimed invention, the lure according to the twelfth or thirteenth aspect may further include an urging member. The urging member urges the weight at least forward. This configuration facilitates returning of the weight toward the head.

[0027] According to the claimed invention, the center of gravity of the lure can be adjusted appropriately. In addition, according to the claimed invention, the center of gravity of the lure can remain stable while the lure is moving through the water.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. 1 is an external perspective view of a lure according to a first embodiment of the claimed invention.

[0029] FIG. 2 is a side view illustrating the internal structure of a lure when a weight is located at the upper portion.

[0030] FIG. 3 is a side view illustrating the internal structure of a lure when the weight is located at the lower portion.

[0031] FIG. 4 is a side view illustrating a modification of the first embodiment.

[0032] FIG. 5 is an external perspective view of a lure according to a second embodiment of the claimed invention.

[0033] FIG. 6 is a perspective view illustrating the internal structure of a lure.

[0034] FIG. 7 is a side view illustrating the internal structure of a lure when a weight is disengaged

from a restricting portion.

[0035] FIG. **8** is a side view illustrating the internal structure of a lure when a weight is engaged with a restricting portion.

[0036] FIG. **9** is a side view illustrating a modification of the first embodiment.

[0037] FIG. **10** is an external perspective view of a lure according to a third embodiment of the claimed invention.

[0038] FIG. **11** is a perspective view illustrating the internal structure of a lure.

[0039] FIG. **12** is a side view illustrating the internal structure of a lure.

[0040] FIG. **13A** is a partially-enlarged side view illustrating the positional relationship between a sliding portion and a restricting portion when a weight is positioned on the head side of a lure.

[0041] FIG. **13B** is a partially-enlarged side view illustrating the positional relationship between a sliding portion and a restricting portion when a weight is positioned on the head side of a lure.

## DESCRIPTION OF EMBODIMENTS

### First Embodiment

[0042] In the first embodiment, the lure according to the claimed invention will be described with reference to a lure **1** in FIG. **1**. As illustrated in FIG. **2**, the lure **1** includes a body **3**, a support **5**, a weight **7**, and an urging member **11**.

#### (Body)

[0043] As illustrated in FIGS. **2** and **3**, the body **3** includes a main body **13**, a first slot **17**, a support hole **19** (see FIG. **3**), and a second slot **22**. The main body **13** includes a head **13a** in a front portion and a tail **13b** in a rear portion.

[0044] As illustrated in FIG. **1**, the main body **13** has a streamlined outer shape that is elongated from the front to the rear. In the lure **1**, where a fishing line (not shown) is connected to a line eye **21**, the direction in which the lure **1** moves when pulled by the fishing line is referred to as “front” and “forward.”

[0045] The direction toward the water surface when the lure **1** moves forward in the water is referred to as “upward”. The direction toward the water bottom is referred to as “downward”. The left-right direction is perpendicular to the front-rear direction and the up-down direction. A transverse direction intersects the front-rear direction. In this embodiment, the up-down direction may also be referred to as the transverse direction.

[0046] As illustrated in FIG. **1**, the main body **13** includes a first main body **29** and a second main body **31**. The first main body **29** and the second main body **31** are defined by dividing the main body **13** along a plane defined in the front-rear and the up-down directions.

[0047] As illustrated in FIGS. **2** and **3**, the first main body **29** includes a first joint surface **29a**. The first joint surface **29a** extends in the front-rear direction. A first front-wall **23a** is positioned on the inner surface at the head **13a** side of the first main body **29**. The first front-wall **23a** extends in the up-down direction. A first rear-wall **25a** is positioned on the inner lateral surface at the tail **13b** side of the first main body **29**. The first rear-wall **25a** extends in the up-down direction.

[0048] A first positioning wall **27a** is positioned on the inner lateral surface at the tail **13b** side of the first body **29**. The first positioning wall **27a** is arranged between the first front-wall **23a** and the first rear-wall **25a** in the front-rear direction. The first positioning wall **27a** extends in the up-down direction. A first upper-rib **15a** is positioned on the inner lateral surface of the first body **29**. The first upper-rib **15a** extends in the front-rear direction above a hook-eye storage **41**.

[0049] A first lower-rib **16a** is positioned on the inner surface of the first body **29**. The first lower-rib **16a** is arranged between the first upper-rib **15a** and the hook-eye storage **41** in the up-down direction. The first lower-rib **16a** extends in the front-rear direction above the hook-eye storage **41**.

[0050] As illustrated in FIG. **1**, the second main body **31** is positioned opposite the first body **29**. The second main body **31** is configured as a mirror image of the first body **29**. In other words, the structure of the second main body **31** is substantially identical to that of the first body **29**. For this reason, elements of the second main body **31** corresponding to those of the first main body **29** are

denoted with the symbol “b.” In each figure, these symbols are written in parentheses for the elements of the second main body **31**.

[0051] For example, the second main body **31** includes a second front-wall **23b**, a second rear-wall **25b**, and a second alignment-wall **27b**. The first front-wall **23a** and the second front-wall **23b** form the front wall **23**. The first rear-wall **25a** and the second rear-wall **25b** form the rear wall **25**. The first alignment-wall **27a** and the second alignment-wall **27b** form the alignment portion **27**. The first upper-rib **15a** and the second upper-rib **15b** form the upper rib **15**.

[0052] As illustrated in FIGS. **2** and **3**, the first joint surface **29a** of the first main body **29** and the joint surface **29b** of the second main body **31** are joined, so that an internal space is formed between the first main body **29** and the second main body **31**. A first slot **17** is formed between the first front-wall **23a** and the second front-wall **23b**.

[0053] The first slot **17** is situated inside the main body **13** on the head **13a** side. The support **5** is inserted through the first slot **17**. The first slot **17** guides the front end of the support **5** in the transverse direction. Specifically, the first slot **17** guides the front end of the support **5** in the up-down direction. The length of the first slot **17** in the up-down direction determines the oscillation range of the support **5**.

[0054] A support hole **19** is formed between the first rear-wall **25a** and the second rear-wall **25b**. The rear end of the support **5** is inserted into the support hole **19**. The support hole **19** holds the rear end of the support **5**, enabling it to oscillate. In this embodiment, the position of the support hole **19** determines the oscillation center (C).

[0055] As illustrated in FIGS. **2** and **3**, a second slot **22** is formed between the first alignment-wall **27a** and the second alignment-wall **27b**. The second slot **22** is situated inside the main body **13** on the tail **13b** side. The rear portion of the support **5** and the rear portion of the urging member **11** are inserted through the second slot **22**. The second slot **22** guides the rear portion of the support **5** and the rear portion of the urging member **11** in the transverse direction. Specifically, the second slot **22** guides the rear end of the support **5** and the rear portion of the urging member **11** in the up-down direction.

[0056] The alignment portion **27** is positioned inside the main body **13** on the tail **13b** side. When the first main body **29** and the second main body **31** are joined, the alignment portion **27** is formed by the first alignment-wall **27a** of the first main body **29** and the second alignment-wall **27b** of the second main body **31**. When the weight **7** moves rearward, the rear portion of the weight **7** abuts against the alignment portion **27**, which causes the weight **7** to be aligned on the tail **13b** side.

(Support)

[0057] As illustrated in FIGS. **2** and **3**, the support **5** supports the weight **7**. The support **5** extends in the front-rear direction inside the main body **13**. The support **5** is arranged inside the main body **13**. Specifically, the support **5** is an axial member with a shaft-like shape.

[0058] As illustrated in FIGS. **3** and **4**, the support **5** is supported by the main body **13** so as to be movable in the transverse direction. In this embodiment, the support **5** is supported by the main body **13** so that the tail end of the support **5** acts as the oscillation center C in the state where the support **5** supports the weight **7**.

[0059] Specifically, the front end of the support **5** is inserted through the first slot **17**. The rear portion of the support **5** is inserted through the second slot **22**. The rear end of the support **5** is inserted into the support hole **19** to engage with either the first rear-wall **25a** or the second rear-wall **25b**.

[0060] The front end of the support **5** and the rear portion of the support **5** move in the up-down direction within the first slot **17** and the second slot **22**, respectively, in response to the position of the weight **7**. The rear end of the support **5** is supported by the support hole **19** so as to oscillate around the position C of the support hole **19**. Consequently, the support **5** oscillates around the rear end of the support **5** as the oscillation center C.

(Weight)

[0061] As illustrated in FIGS. 2 and 3, the weight 7 is supported by the support 5 inside the main body 13. As illustrated in FIG. 3, the weight 7 is supported by the support 5 so as to be movable in the front-rear direction.

[0062] The weight 7 includes a weight body 7a and a hole 7b. The hole 7b is formed through the weight body 7a. The hole 7b penetrates the weight body 7a in the front-rear direction. The support 5 is inserted through the hole 7b. With this configuration, the weight body 7a moves along the support 5 in the front-rear direction as illustrated in FIG. 2.

[0063] The weight 7 moves in the front-rear direction between a first position on the head 13a side (indicated by a solid line in FIG. 2) and a second position on the tail 13b side (indicated by a two-dot dashed line in FIG. 2). The first position on the head 13a side is where the front portion of the weight 7 most approaches the front wall 23. In this embodiment, the first position on the head 13a side is specifically where the front portion of the weight 7 abuts against the front wall 23.

[0064] The second position on the tail 13b side is where the rear portion of the weight 7 most approaches the alignment portion 27. In this embodiment, the second position on the tail 13b side is specifically where the rear portion of the weight 7 abuts against the alignment portion 27.

[0065] As illustrated in FIGS. 2 and 3, the weight 7 is supported by the support 5 so as to be movable in the up-down direction. Specifically, when the support 5 oscillates around the oscillation center C, the weight 7 moves in the up-down direction while being supported by the support 5. In this state, the weight 7 moves along the support 5 in the front-rear direction.

(Urging Member)

[0066] The urging member 11 urges the weight 7 at least forward. In this embodiment, as illustrated in FIGS. 2 and 3, the urging member 11 is a coil spring. The urging member 11 is arranged between the weight 7 and the rear wall 25. When the weight 7 is positioned at the first position on the head 13a side, the urging member 11 is compressed.

[0067] In the lure 1 with the above configuration, as illustrated in FIG. 2, the weight 7, while being supported by the support 5, moves along the support 5 in the front-rear direction between the front and rear portions of the body 3. Specifically, the weight 7 moves along the support 5 in the front-rear direction between the first position on the head 13a side (indicated by the solid line in FIG. 2) and the second position on the tail 13b side (indicated by the two-dot dashed line in FIG. 2).

[0068] When a downward force is applied to the weight 7 in the state where the weight 7 is positioned at the solid-line position in FIG. 2 (the first position on the head 13a side), the support 5 oscillates downward as illustrated in FIG. 3, and the weight 7 moves downward.

[0069] When the weight 7 is positioned at the solid-line position in FIG. 3 (the first position on the head 13a side), the weight 7 moves along the support 5 in the front-rear direction. In this state, when an upward force is applied to the weight 7, the support 5 oscillates upward, and the weight 7 moves upward. As a result, the weight 7 returns to the solid-line position in FIG. 2 (the first position on the head 13a side).

[0070] When the weight 7 is positioned between the first position on the head 13a side and the second position on the tail 13b side in the front-rear direction, and a force is applied to the weight 7 in the upward or downward direction, the support 5 oscillates upward or downward. That is, the weight 7 moves upward or downward at a position between the first position on the head 13a side and the second position on the tail 13b side.

[0071] In the lure 1, the support 5 is movable in the transverse direction, and the weight 7 is supported by the support 5 so as to be movable in the front-rear direction. Therefore, when the support 5 moves in the transverse direction, the weight 7 also moves in the transverse direction together with the support 5. This configuration enhances the degree of freedom in the movement path of the weight 7 in the lure 1 compared to the prior art. That is, in the lure 1, the center of gravity position can be adjusted suitably. Additionally, the weight 7 moves not only in the forward and backward directions but also in the transverse direction in response to the action (e.g., winding operation) imparted to the lure 1 by the angler. This results in more irregular behavior of the lure 1.

[0072] In the lure **1**, the weight **7** supported by the support **5** can oscillate, thereby enhancing the degree of freedom of the movement path of the weight **7**. In the lure **1**, the front end of the support **5** with a shaft-like shape is guided in the transverse direction by the first slot **17** of the body **3**. Therefore, the weight **7** supported by the support **5** can move stably in the transverse direction. Furthermore, in the lure **1**, the weight **7** can easily return to the head **13a** side.

#### Modifications of First Embodiment

[0073] One embodiment of the claimed invention has been described above. However, the claimed invention is not limited to this first embodiment, and various modifications are possible without departing from the scope of the claimed invention.

[0074] In the configuration of the first embodiment, the lure **1** may further include a magnet **10**, as illustrated in FIG. **4**. In this case, the weight **7** is at least partially comprised of magnetic material. In the first embodiment, the weight **7** is comprised of magnetic material. With this configuration, the weight **7** is attracted to the magnet **10**.

[0075] The magnet **10** is positioned on the main body **13** to hold the weight **7** in the first position on the head **13a** side. In the first embodiment, the magnet **10** is located on the upper rib **15**. With this configuration, when the weight **7** is placed at the position indicated by the solid line in FIG. **4**, corresponding to the first position on the head **13a** side, the weight **7** is held by the magnet **10**. In this state, when the backward force applied to the weight **7** exceeds the magnetic attraction of the magnet **1** for the weight **7**, the weight **7** moves backward along the support **5**.

[0076] When the weight **7** is positioned at the first position on the head **13a** side as illustrated by the solid line in FIG. **4**, and the downward force applied to the weight **7** exceeds the magnetic attraction of the magnet **1** for the weight **7**, the support **5** oscillates downward and the weight **7** moves downward.

#### Second Embodiment

[0077] The lure according to a second embodiment of the claimed invention will be described using the lure **1** illustrated in FIG. **5**. As illustrated in FIG. **6**, the lure **1** includes a body **3**, a support **5**, a weight **7**, a restricting portion **9**, and an urging member **11**. FIG. **6** is a perspective view of the weight **7** engaged with the restricting portion **9**.

#### (Body)

[0078] As illustrated in FIGS. **6** and **7**, the body **3** includes a main body **13**, a first slot **17**, a support hole **19** (see FIG. **7**), and a second slot **22**. The main body **13** includes a head **13a** in a front portion and a tail **13b** in a rear portion.

[0079] As illustrated in FIG. **1**, the main body **13** has a streamlined outer shape that is elongated from the front to the rear. In lure **1**, where a fishing line (not shown) is connected to a line eye **21**, the direction in which lure **1** advances as it is pulled by the fishing line is determined as “front” and “forward”.

[0080] The direction toward the water surface when the lure **1** moves forward in the water is referred to as “upward”. The direction toward the water bottom is referred to as “downward”. The left-right direction is perpendicular to the front-rear direction and the up-down direction. A transverse direction intersects the front-rear direction. In this embodiment, the up-down direction may also be referred to as the transverse direction.

[0081] As illustrated in FIG. **5**, the main body **13** includes a first main body **29** and a second main body **31**. The first main body **29** and the second main body **31** are defined by dividing the main body **13** along a plane defined in the front-rear and the up-down directions. As illustrated in FIGS. **6** and **7**, the first main body **29** includes a first joint surface **29a**. The first joint surface **29a** extends in the front-rear direction. A first front-wall **23a** is positioned on the inner lateral surface of the first main body **29** on the head **13a** side. The first front-wall **23a** extends in the up-down direction. A first rear-wall **25a** is positioned on the inner lateral surface of the first main body **29** on the tail **13b** side. The first rear-wall **25a** extends in the up-down direction.

[0082] A first alignment-wall **27a** is positioned on the inner lateral surface of the first main body **29**



on the tail **13b** side. The first alignment-wall **27a** is arranged between the first front-wall **23a** and the first rear-wall **25a** in the front-rear direction. The first alignment-wall **27a** extends in the up-down direction.

[0083] A first upper-rib **15a** is positioned on the inner surface of the first main body **29**. The first upper-rib **15a** extends in the front-rear direction above the hook-eye storage **41**.

[0084] A first lower-rib **16a** is positioned on the inner surface of the first main body **29**. The first lower-rib **16a** is situated between the first upper-rib **15a** and the hook-eye storage **41** in the up-down direction. The first lower-rib **16a** extends in the front-rear direction above the hook-eye storage **41**.

[0085] A first front-rib **18a** is positioned on the inner surface of the first main body **29**. The first front-rib **18a** is arranged in front of the hook-eye storage **41**. The first front-rib **18a** extends downward from the front of the first lower-rib **16a**.

[0086] As illustrated in FIG. 5, the second main body **31** is positioned opposite the first main body **29**. The second main body **31** is configured as a mirror image of the first main body **29**. In other words, the structure of the second main body **31** is substantially identical to that of the first main body **29**. For this reason, elements of the second main body **31** corresponding to those of the first main body **29** are denoted with the symbol “b.” In each figure, these symbols are written in parentheses for the elements of the second main body **31**.

[0087] For example, the second main body **31** includes a second front-wall **23b**, a second rear-wall **25b**, a second alignment-wall **27b**, a second upper-rib **15b**, a second lower-rib **16b**, and a second front-rib **18b**. The first front-wall **23a** and the second front-wall **23b** form the front wall **23**. The first rear-wall **25a** and the second rear-wall **25b** form the rear wall **25**. The first alignment-wall **27a** and the second alignment-wall **27b** form the alignment portion **27**.

[0088] The first upper-rib **15a** and the second upper-rib **15b** form the upper rib **15**. The first lower-rib **16a** and the second lower-rib **16b** form the lower rib **16**. The first front-rib **18a** and the second front-rib **18b** form the front rib **18**.

[0089] As illustrated in FIGS. 6 and 7, the first joint surface **29a** of the first main body **29** and the joint surface **29b** of the second main body **31** are joined, so that an internal space is formed between the first main body **29** and the second main body **31**. A first slot **17** is formed between the first front-wall **23a** and the second front-wall **23b**.

[0090] The first slot **17** is positioned inside the main body **13** on the head **13a** side. The support **5** is inserted through the first slot **17**. The first slot **17** guides the front end of the support **5** in the transverse direction. Specifically, the first slot **17** guides the front end of the support **5** in the up-down direction. The length of the first slot **17** in the up-down direction determines the oscillation range of the support **5**.

[0091] A support hole **19** is formed between the first rear-wall **25a** and the second rear-wall **25b**. The rear end of the support **5** is inserted into the support hole **19**. The support hole **19** supports the rear end of the support **5**, enabling it to oscillate. In this embodiment, the position of the support hole **19** determines the oscillation center (C).

[0092] As illustrated in FIGS. 6 and 7, a second slot **22** is formed between the first alignment-wall **27a** and the second alignment-wall **27b**. The second slot **22** is positioned inside the main body **13** on the tail **13b** side. The rear portion of the support **5** and the rear portion of the urging member **11** are inserted through the second slot **22**. The second slot **22** guides the rear portion of the support **5** in the transverse direction. Specifically, the second slot **22** guides the rear end of the support **5** in the up-down direction.

[0093] The alignment portion **27** is positioned inside the main body **13** on the tail **13b** side. When the first main body **29** and the second main body **31** are joined, the alignment portion **27** is formed by the first alignment-wall **27a** of the first main body **29** and the second alignment-wall **27b** of the second main body **31**. When the weight **7** moves rearward, the rear portion of the weight **7** abuts against the alignment portion **27**, which causes the weight **7** to be aligned on the tail **13b** side.

[0094] As illustrated in FIGS. 6 and 7, the upper rib 15 is positioned above the weight 7. When the first main body 29 and the second main body 31 are joined, the upper rib 15 is formed above the lower rib 16. The lower rib 16 is situated below the support 5 and the urging member 11. A space is created between the upper rib 15 and the lower rib 16 to allow the weight 7 to pass between the upper rib 15 and the lower rib 16. The front rib 18 extends downward from the front portion of the lower rib 16, in front of the hook-eye storage 41.

(Support)

[0095] As illustrated in FIGS. 6 and 7, the support 5 supports the weight 7. The support 5 extends in the front-rear direction inside the main body 13. The support 5 is disposed inside the main body 13. Specifically, the support 5 is an axial member with a shaft-like shape.

[0096] As illustrated in FIGS. 7 and 8, the support 5 is supported by the main body 13 so as to be movable in the transverse direction. In this embodiment, the support 5 is supported by the main body 13 so that the tail end of the support 5 acts as the oscillation center C in the state where the support 5 supports the weight 7.

[0097] Specifically, the front end of the support 5 is inserted through the first slot 17. The rear portion of the support 5 is inserted through the second slot 22. The rear end of the support 5 is inserted into the support hole 19 to engage with either the first rear-wall 25a or the second rear-wall 25b.

[0098] The front end of the support 5 and the rear portion of the support 5 move in the up-down direction within the first slot 17 and the second slot 22, respectively, in response to the position of the weight 7. The rear end of the support 5 is supported by the support hole 19 so as to oscillate around the position C of the support hole 19. Consequently, the support 5 oscillates around the rear end of the support 5 as the oscillation center C.

(Weight)

[0099] As illustrated in FIGS. 6 and 7, the weight 7 is supported by the support 5 inside the main body 13. As illustrated in FIG. 7, the weight 7 is supported by the support 5 so as to be movable in the front-rear direction. As illustrated in FIGS. 7 and 8, the weight 7 is supported by the support 5 so as to be movable in the up-down direction. As illustrated in FIG. 8, the weight 7 is restricted by the restricting portion 9 as the weight 7 moves downward.

[0100] The weight 7 includes a weight body 7a and a hole 7b. The weight body 7a is arranged between the upper rib 15 and the lower inner surface of the main body 13 in the up-down direction. A hole 7b is formed through the weight body 7a. The hole 7b penetrates the weight body 7a in the front-rear direction. The support 5 is inserted through the hole 7b. With this configuration, as illustrated in FIG. 7, the weight body 7a moves along the support 5 in the front-rear direction.

[0101] The weight 7 moves in the front-rear direction between a first position on the head 13a side (indicated by a solid line in FIG. 2) and a second position on the tail 13b side (indicated by a two-dot dashed line in FIG. 2). The first position on the head 13a side is where the front portion of the weight 7 most approaches the front wall 23. In this embodiment, the first position on the head 13a side is specifically where the front portion of the weight 7 abuts against the front wall 23.

[0102] The second position on the tail 13b side is where the rear portion of the weight 7 most approaches the alignment portion 27. In this embodiment, the second position on the tail 13b side is specifically where the rear portion of the weight 7 abuts against the alignment portion 27.

[0103] When a downward force is applied to the weight 7 in a state in which the weight 7 is close to the upper rib 15 at the first position on the head 13a side as illustrated in FIG. 7, the support 5 oscillates downward as illustrated in FIG. 8. This causes the weight 7 to engage with the restricting portion 9. Similarly, when an upward force is applied to the weight 7 in the state illustrated in FIG. 8, the support 5 oscillates upward as illustrated in FIG. 7. This causes the weight 7 to disengage from the restricting portion 9.

(Restricting Portion)

[0104] As illustrated in FIGS. 6, 7 and 8, the restricting portion 9 is configured to limit the weight 7

from moving rearward. The restricting portion **9** is positioned on the inner surface of the main body **13** in the front portion of the main body **13**. In this embodiment, as illustrated in FIGS. **7** and **8**, the restricting portion **9** is a recess **9a** formed on the inner surface of the main body **13**. The recess **9a** is located in the front and lower portion of the main body **13**. Specifically, the recess **9a** has a concave shape in which the weight **7** can be situated. The recess **9a** is formed by the front wall **23**, the lower inner surface **13c** of the main body **13**, the inner lateral surfaces **13d** of the main body **13**, and the front rib **18**.

(Urging Member)

[0105] The urging member **11** urges the weight **7** at least forward. In this embodiment, as illustrated in FIGS. **6**, **7**, and **8**, the urging member **11** is a coil spring. The urging member **11** is arranged between the weight **7** and the rear wall **25**. When the weight **7** is positioned at the first position on the head **13a** side, the urging member **11** is compressed.

[0106] In the lure **1** with the above configuration, as illustrated in FIG. **7**, the weight **7**, while being supported by the support **5**, moves along the support **5** in the front-rear direction between the front and rear portions of the body **3**. Specifically, the weight **7** moves along the support **5** in the front-rear direction between the first position on the head **13a** side (indicated by the solid line in FIG. **7**) and the second position on the tail **13b** side (indicated by the two-dot dashed line in FIG. **7**).

[0107] When a downward force is applied to the weight **7** positioned at the first position on the head **13a** side, as illustrated in FIG. **7**, the weight **7** moves downward, as illustrated in FIG. **8**. As a result, as illustrated in FIGS. **6** and **8**, the restricting portion **9** restricts the weight **7** from moving rearward. Thus, the restricting portion **9** in the front portion of the body **3** easily limits the rearward movement of the weight **7**. In other words, in lure **1**, a simple structure stabilizes the center of gravity while the lure is moving through the water.

[0108] In lure **1**, the weight **7** engages with the recess **9a** which functions as the restricting portion **9**. As a result, the rearward movement of the weight **7** is reliably restricted. Furthermore, in lure **1**, the support **5** has a shaft-like shape and oscillates around the tail side as the oscillation center. This configuration allows the weight **7** to move easily in the up-and-down direction. Additionally, in lure **1**, the urging member **11** urges the weight **7** at least forward, enabling the weight **7** to be easily returned to the head **13a** side.

Modification of Second Embodiment

[0109] One embodiment of the claimed invention has been described above. However, the claimed invention is not limited to the second embodiment, and various modifications are possible within the scope of the invention.

[0110] In the configuration of the second embodiment, as illustrated in FIG. **9**, the lure **1** may further include a magnet **10**. In this case, at least part of the weight **7** is comprised of a magnetic material. In the second embodiment, the weight **7** is comprised of a magnetic material. As a result, the weight **7** is attracted to the magnet **10**.

[0111] The magnet **10** is positioned on the main body **13** so that the support **5** is arranged between the recess **9a** and the magnet **10** in the up-down direction. Specifically, the magnet **10** is positioned on the upper rib **15** of the main body **13** above the recess **9a**. In this configuration, when the lure **1** is suspended with its head-side facing upward, the recess **9a** remains disengaged due to the magnetic force of the magnet **10**. As a result, the weight **7** moves smoothly toward the tail during casting.

Third Embodiment

[0112] The configuration of lure **1** in a third embodiment is substantially the same as that of the second embodiment, except for the configurations of the body **3**, the restricting portion **9**, and the weight **7**. In the third embodiment, components identical to those in the second embodiment are denoted by the same reference numerals. Descriptions of components similar to those in the second embodiment are omitted from the third embodiment. Omitted descriptions in the third embodiment are based on the descriptions provided in the second embodiment.

[0113] The lure **101** of the third embodiment is depicted in FIG. **10**. As illustrated in FIG. **11**, the lure **101** includes a body **103**, a support **5**, a weight **107**, a restricting portion **109**, and an urging member **11**. The lure **101** illustrated in FIG. **11** is a perspective view where the weight **107** is engaged with the restricting portion **109**. The configurations of the support **5** and the urging member **11** are each substantially the same as those in the second embodiment.

(Body)

[0114] As illustrated in FIGS. **11** and **12**, the body **103** includes a main body **113**, a first support hole **117**, a second support hole **19**, and an insertion hole **122**. The first support hole **117** is located in the main body **113**. The first support hole **117** is formed when the first joint surface **29a** of the first main body **29** and the joint surface **29b** of the second main body **31** are joined, similar to the first slot **17** of the second embodiment. The front portion of the support **5** is inserted into the first support hole **117**.

[0115] The configuration of the second support hole **19** is the same as in the second embodiment. The front end of the support **5** is fixed to the main body **113**, for example, the first hook-eye storage **42** of the main body **113**. The rear end of the support **5** is inserted into the support hole **19** and engaged with either the first rear-wall **25a** or the second rear-wall **25b**. In the third embodiment, the support **5**, extending in the front-rear direction, does not oscillate as it does in the second embodiment.

[0116] The insertion hole **122** is positioned in the main body **113**. The insertion hole **122** is formed when the first joint surface **29a** of the first main body **29** and the joint surface **29b** of the second main body **31** are joined, similar to the second slot portion **22** of the second embodiment. The rear portion of the support **5** and the rear portion of the urging member **11** are inserted into the insertion hole **122**.

(Weight)

[0117] As illustrated in FIG. **11**, the weight **107** has a weight body **57** and a sliding portion **67**. The weight body **57** is arranged between the upper rib **15** and the lower inner surface of the main body **113** in the up-down direction. The weight body **57** includes a first body **57a** and a first hole **57b**. The first body **57a** is arranged between the upper rib **15** and the lower inner surface of the main body **113** in the up-down direction. The first hole **57b** is formed through the first body **57a**. The first hole **57b** penetrates the first body **57a** in the front-rear direction. The support **5** is inserted through the first hole **57b**. With this configuration, the weight body **57** moves along the support **5** in the front-rear direction.

[0118] The sliding portion **67** is arranged between the upper rib **15** and the lower inner surface of the main body **113** in the up-down direction. The sliding portion **67** is positioned adjacent to the weight body **57** in the front-rear direction. The sliding portion **67** is positioned behind the weight body **57**. The sliding portion **67** is arranged between the weight body **57** and the urging member **11**. Specifically, the front portion of the sliding portion **67** is positioned adjacent to the rear portion of the weight body **57**. A washer **77** is placed on the rear portion of the sliding portion **67**. The urging member **11** contacts with the washer **77**.

[0119] As illustrated in FIGS. **12**, **13A**, and **13B**, the sliding portion **67** is configured to slide with the restricting portion **109**. The sliding portion **67** is preferably comprised of synthetic resin with high sliding properties or metal with high sliding properties. The sliding portion **67** includes a second body **67a** and a second hole **67b**. The second body **67a** includes a lower surface **67a1** and a contact surface **67a2**. The lower surface **67a1** extends in the front-rear direction. The contact surface **67a2** extends upward from the lower surface **67a1** rearward. The contact surface **67a2** contacts the restricting portion **109**.

[0120] As illustrated in FIGS. **13A** and **13B**, the second hole **67b** is located in the second body **67a**. The second hole **67b** penetrates the second body **67a** in the front-rear direction. The shaft-shaped support **5** is inserted through the second hole **67b**. With this configuration, the weight body **57** and the sliding portion **67** move along the support **5** in the front-rear direction.

[0121] As illustrated in FIGS. 13A and 13B, the second hole **67b** has an inner diameter greater than the inner diameter of the first hole **57b**. The inner diameter of the second hole **67b** is greater than the outer diameter of the support **5**. As illustrated in FIG. 13A, when the weight **7** is positioned at the first position on the head **13a** side, the contact surface **67a2** of the second body **67a** is contact with the restricting portion **109**. In this case, the lower surface **67a1** of the second body **67a** is spaced apart from the restricting portion **109** in the up-down direction. In this state, the support **5** is positioned upward within the second hole **67b**.

[0122] As illustrated in FIG. 13B, when the weight **107** is positioned at the second position on the tail **13b** side, the lower surface **67a1** of the second body **67a** is contact with the first lower surface **113a** of the main body **113**. In this state, the support **5** is positioned downward within the second hole **67b**.

(Restricting Portion)

[0123] As illustrated in FIGS. 12, 13A, and 13B, the restricting portion **109** is positioned on the inner surface of the main body **113**. The restricting portion **109** includes an inclined plane **109a** (an example of a slope). The inclined plane **109a** restricts the weight **107** from moving rearward through sliding resistance against the sliding portion **67**.

[0124] The inclined plane **109a** is arranged between the first lower surface **113a** of the main body **113** and the second lower surface **113b** of the main body **113** in the front-rear direction. The first lower surface **113a** is formed by the upper surface of the second hook-eye storage **41** and extends in the front-rear direction.

[0125] The second lower surface **113b** also extends in the front-rear direction and is positioned in front of the first lower surface **113a**. The rear end of the second lower surface **113b** is located below the front end of the first lower surface **113a**. The inclined plane **109a** extends upward from the front end of the second lower surface **113b** toward the rear end of the first lower surface **113a**.

[0126] In other words, the inclined plane **109a** inclines from the front end of the first lower surface **113a** toward the outer surface of the main body **113**. In this embodiment, the inclined plane **109a** is formed by the front surface of the second hook-eye storage **41**.

[0127] In the lure **101** with the above configuration, the weight **107** moves along the support **5** in the front-rear direction between the front and rear portions of the body **103** while being supported by the support **5**. Specifically, the weight **107** moves along the support **5** in the front-rear direction between a first position on the head **13a** side (see the solid line in FIG. 12) and a second position on the tail **13b** side (see the two-dot dashed line in FIG. 12).

[0128] As illustrated in FIG. 13A, when the weight **107** is at the first position on the head **13a** side, the contact surface **67a2** of the sliding portion **67** of the weight **107** is in contact with the inclined plane **109a** of the restricting portion **109**. In this state, when a backward force is applied to the weight **107**, the backward movement of the weight **107** is restricted through sliding resistance between the contact surface **67a2** of the sliding portion **67** and the inclined plane **109a** of the restricting portion **109**.

[0129] Thus, in lure **1**, the restricting portion **109** on the body **3** facilitates restriction of the backward movement of the weight **107**. In other words, in lure **1**, a simple structure stabilizes the center of gravity while the lure is moving through the water.

[0130] In addition, the sliding portion **67** of lure **101** includes a second body **67a** and a second hole **67b**. The second hole **67b** has an inner diameter greater than the outer diameter of the shaft-shaped support **5**. With this configuration, where the inner diameter of the second hole **67b** is greater than the outer diameter of the support, the sliding portion **67** can move easily in the transverse direction. The second hole **67b** may be an elongated hole extending in the transverse direction. For example, the elongated hole may extend in the up-down direction. The elongated hole preferably has an elliptical cross section.

#### REFERENCE SIGNS LIST

[0131] **1**, **101** lure [0132] **3**, **103** Body [0133] **5** Support [0134] **7**, **107** Weight [0135] **9**, **109**

Restricting portion [0136] **109a** Inclined plane [0137] **9a** Recess [0138] **10** Magnet [0139] **11**  
Urging member [0140] **13**, **113** Main body [0141] **13a** Head [0142] **13b** Tail [0143] **17** First slot  
[0144] **57** Weight body [0145] **67** Sliding portion [0146] **67a** Second body of sliding portion [0147]  
**67b** Second hole in sliding portion

## Claims

1. A lure, comprising: a body including a head in a front portion and a tail in a rear portion; a support extending in a front-rear direction inside the body; and a weight supported by the support so as to be movable in the front-rear direction, wherein at least part of the weight is movable in a transverse direction that intersects the front-rear direction.
2. The lure according to claim 1, wherein the support is supported by the body so as to be movable in the transverse direction.
3. The lure according to claim 2, wherein the support is an axial member with a shaft-like shape, and the axial member oscillates around the tail side as an oscillation center.
4. The lure according to claim 3, wherein the axial member includes a front-end portion on the head side and a rear-end portion on the tail side, the rear-end portion being supported by the body to be able to oscillate, and the body has a groove for guiding the front-end portion in the transverse direction.
5. The lure according to claim 2, further comprising an urging member for urging the weight at least forward.
6. The lure according to claim 1, further comprising: a restricting portion positioned on an inner surface of the body in the front portion of the body and configured to limit the weight from moving rearward, wherein the weight is restricted by the restricting portion when at least part of the weight moves in the transverse direction.
7. The lure according to claim 6, wherein the support is supported by the body so as to be movable in the transverse direction, and the weight is restricted by the restricting portion when the support moves in the transverse direction.
8. The lure according to claim 7, wherein the restricting portion is a recess formed on the inner surface of the body.
9. The lure according to claim 8, further comprising: a magnet positioned on the body so as to face the recess with the support in between, wherein the weight is at least partially comprised of magnetic material.
10. The lure according to claim 9, wherein the support is an axial member with a shaft-like shape, and the axial member oscillates around the tail side as an oscillation center.
11. The lure according to claim 7, further comprising an urging member for urging the weight at least forward.
12. The lure according to claim 6, wherein the weight includes a weight body and a sliding portion, the restricting portion includes a slope that inclines toward an outer surface of the body, and when at least part of the weight moves in the transverse direction, the sliding portion contacts the slope, and the slope restricts the weight from moving rearward through sliding resistance against the sliding portion.
13. The lure according to claim 12, wherein the support is an axial member with a shaft-like shape, the sliding portion includes a main body and a hole in the main body through which the axial member is inserted, and the hole has an inner diameter greater than an outer diameter of the axial member.
14. The lure according to claim 12, further comprising an urging member for urging the weight at least forward.
15. The lure according to claim 8, further comprising an urging member for urging the weight at least forward.

**16.** The lure according to claim 13, further comprising an urging member for urging the weight at least forward.

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